ClinicQueue API Specification

A. User Registration

API Endpoint	/api/auth/register	
method	POST	
Request Parameters	**Type**	**Description**
fullName	string	User's full name
email	string	User's email address
password	string	User's password (the server hashes it)
affiliation	string	student, faculty and staff
Register a new ClinicQueue user account		
Success		
Status Code	201	
Response Type	json	
Response Data	**Type**	**Description**
message	string	Registration success message
userId	integer	Unique ID of the created user
Sample Response		

```
{
  "message": "User registered successfully",
  "userId": 101
}
```

Fail		
Status Code	400	
Response Type	json	
Response Data	**Type**	**Description**
error	string	Validation message why the request is failed

Sample Response		

{
"error": "Invalid registration information"
}		

Status Code	409	
Response Type	json	
Response Data	**Type**	**Description**
error	string	Email is already registered
Sample Response		

{
"error": "Email is already registered"
}		

B. User Login

API Endpoint	/api/auth/login	
method	POST	
Request Parameters	**Type**	**Description**
email	string	Registered user's email address
password	string	User Password
Validates the registered user		
Success		
Status Code	200	
Response Type	json	
Response Data	**Type**	**Description**
message	string	Login Success Message
userId	integer	ID of the registered user
role	string	Access role of the user

Sample Response		
<pre>{ "message": "Login successful", "userId": 101, "role": "patient" }</pre>		
Fail		
Status Code	401	
Response Type	json	
Response Data	**Type**	**Description**
error	string	Invalid Credentials
Sample Response		
<pre>{ "error": "Invalid email or password" }</pre>		

C. Check User Profile

API Endpoint	/api/users/{id}	
method	GET	
Request Parameters	**Type**	**Description**
id	integer	Required. Path parameter. Unique ID of the user
Retrieves the user's profile information		
Success		
Status Code	200	
Response Type	json	
Response Data	**Type**	**Description**
id	integer	Unique user ID
fullName	string	User's full name
email	string	User's email address

affiliation	string	User's clinic affiliation
role	string	User's access role
isActive	boolean	Indicates whether the account is active
Sample Response		
<pre>{ "id": 101, "fullName": "John Albert Masangkay", "email": "johnalbert@gmail.com", "affiliation": "student", "role": "patient", "isActive": true }</pre>		
Fail		
Status Code	404	
Response Type	json	
Response Data	**Type**	**Description**
error	string	User not registered
Sample Response		
<pre>{ "error": "User not found" }</pre>		

D. User Logout

API Endpoint	/api/auth/logout	
method	POST	
Request Parameters	**Type**	**Description**
none	none	none
Ends the session. The server invalidates the token and the frontend clears its stored copy.		
Success		
Status Code	200	

Response Type	json	
Response Data	**Type**	**Description**
message	string	Logout Success Message
Sample Response		

{
"message": "Logged out successfully"
}		

Fail		
Status Code	401	
Response Type	json	
Response Data	**Type**	**Description**
error	string	Missing, invalid, or expired authentication token
Sample Response		

{
"error": "Unauthorized: invalid or expired token"
}		

CLINIC SLOTS

E. List Slots

API Endpoint	/api/slots	
method	GET	
Request Parameters	**Type**	**Description**
from	date	Optional. Query parameter. Default is today
to	date	Optional. Query parameter. Default is 7 days after from
Lists clinic slots. Patients see only slots where isBookable is true and remainingCapacity is above 0. Staff see all slots.		

Success		
Status Code	200	
Response Type	json	
Response Data	**Type**	**Description**
slots	array of Slot	Slots in the date range
Sample Response		


```
{
  "slots": [
    {
      "id": 12,
      "clinicDate": "2026-11-12",
      "startTime": "10:15",
      "endTime": "10:45",
      "capacity": 2,
      "remainingCapacity": 1,
      "isBookable": true
    }
  ]
}
```


Fail		
Status Code	400	
Response Type	json	
Response Data	**Type**	**Description**
error	string	to is before from or a date is invalid
Sample Response		


```
{
   "error": "Invalid date range"
}
```


F. Create Slot

API Endpoint	/api/slots	
method	POST	
Request Parameters	**Type**	**Description**

clinicDate	date	Required. Cannot be in the past
startTime	time	Required. Slot start
endTime	time	Required. Must be after startTime
capacity	integer	Required. At least 1
isBookable	boolean	Optional. Default is true
Creates an available clinic slot.		
Success		
Status Code	201	
Response Type	json	
Response Data	**Type**	**Description**
message	string	Slot created message
id	integer	Unique ID of the new slot
Sample Response		


```
{
  "message": "Slot created successfully",
  "id": 12
}
```


Fail		
Status Code	400	
Response Type	json	
Response Data	**Type**	**Description**
error	string	Missing or invalid slot details
Sample Response		


```
{
   "error": "Invalid slot details"
}
```


Status Code	409
Response Type	json

Response Data	Type	Description
error	string	Overlaps an existing slot
Sample Response		

{
"error": "Slot overlaps an existing slot"
}		

G. Update Slot

API Endpoint	/api/slots/{id}	
method	PATCH	
Request Parameters	**Type**	**Description**
id	integer	Required. Path parameter. Unique ID of the slot
clinicDate	date	Optional. New clinic date
startTime	time	Optional. New start time
endTime	time	Optional. New end time
capacity	integer	Optional. Cannot be lower than the number already booked
isBookable	boolean	Optional. Open or close the slot for booking
Updates a slot. Send only the fields to change.		
Success		
Status Code	200	
Response Type	json	
Response Data	**Type**	**Description**
message	string	Slot updated message
Sample Response		
{ "message": "Slot updated successfully"		

}		
Fail		
Status Code	400	
Response Type	json	
Response Data	**Type**	**Description**
error	string	Invalid slot details
Sample Response		

{
"error": "Invalid slot details"
}		

Status Code	404	
Response Type	json	
Response Data	**Type**	**Description**
error	string	No slot with that ID
Sample Response		

{
"error": "Slot not found"
}		

Status Code	409	
Response Type	json	
Response Data	**Type**	**Description**
error	string	Capacity below booked count
Sample Response		

{
"error": "Capacity cannot be lower than booked appointments"
}		

Status Code	409	
Response Type	json	
Response Data	**Type**	**Description**

error	string	Overlaps an existing slot
Sample Response		

{
"error": "Slot overlaps an existing slot"
}		

H. Delete Slot

API Endpoint	/api/slots/{id}	
method	DELETE	
Request Parameters	**Type**	**Description**
id	integer	Required. Path parameter. Unique ID of the slot
Deletes a slot that has no active appointments.		
Success		
Status Code	200	
Response Type	json	
Response Data	**Type**	**Description**
message	string	Slot deleted message
Sample Response		

{
"message": "Slot deleted successfully"
}		

Fail		
Status Code	404	
Response Type	json	
Response Data	**Type**	**Description**
error	string	No slot with that ID
Sample Response		
{ "error": "Slot not found"		

}		
Status Code	409	
Response Type	json	
Response Data	**Type**	**Description**
error	string	Slot still has active appointments
Sample Response		

{
"error": "Slot has active appointments. Set isBookable to false instead"
}		

I. Join Walk-In Queue

API Endpoint	/api/v1/queues/{queue_id}/walk-in	
method	POST	
Request Parameters	**Type**	**Description**
queue_id	string (UUID)	Required. Path parameter. Unique identifier of the target queue session.
patient_id	string (UUID)	Optional. Body parameter. Registered patient identifier if existing account exists.
first_name	string	Required. Body parameter. Patient given name.
last_name	string	Required. Body parameter. Patient family name.
phone_number	string	Required. Body parameter. E.164 formatted contact phone number.

reason_for_visit	string	Optional. Body parameter. Brief explanation of consultation request.
priority	string	Optional. Body parameter. NORMAL or EMERGENCY.
Registers a walk-in patient into an active queue, assigns a sequential tracking token, and calculates an initial estimated wait time.		
Sample Response		
<pre>{ "first_name": "Jane", "last_name": "Doe", "phone_number": "+639171234567", "reason_for_visit": "Fever and Headache", "priority": "NORMAL" }</pre>		
Success		
Status Code	201	
Response Type	json	
Response Data	**Type**	**Description**
queue_entry_id	string (UUID)	Unique identifier of created queue entry
queue_id	string (UUID)	Target queue session identifier
ticket_number	string	Sequential tracking ticket number
position_in_queue	integer	Current position relative to waiting list
estimated_wait_minutes	integer	Estimated wait duration in minutes
status	string	Queue entry state (e.g. WAITING)
priority	string	NORMAL or EMERGENCY
tracking_token	string	Secret token a walk-in without an account sends

		in the X-Tracking-Token header to view this entry,
joined_at	string (ISO 8601)	Timestamp when entry was added
Sample Response		

{
"queue_entry_id": "qentry_987f6543-e21b-12d3-a456-426614174000",
"queue_id": "q_123e4567-e89b-12d3-a456-426614174000",
"ticket_number": "A-042",
"position_in_queue": 5,
"estimated_wait_minutes": 25,
"status": "WAITING",
“priority”: “NORMAL”,
“Tracking_token”: “trk_8f3alc9e72b4”
"joined_at": "2026-10-06T22:10:00Z"
}		

Fail		
Status Code	400 / 404 / 409	
Response Type	json	
Response Data	Type	Description
error_code	string	Application-specific error identifier (e.g. QUEUE_CLOSED, INVALID_INPUT)
message	string	Human-readable description of the error
Sample Response		

{
"error_code": "QUEUE_CLOSED",
"message": "The specified queue session is no longer accepting walk-in registrations."
}		

J. Get Queue Status

API Endpoint	/api/v1/queues/{queue_id}/status
method	GET

Request Parameters	Type	Description
queue_id	string (UUID)	Required. Path parameter. Unique identifier of the queue session.
Retrieves real-time operational status for a specific queue, including current serving numbers, total waiting patients, and average wait metrics.		
Success		
Status Code	200	
Response Type	json	
Response Data	**Type**	**Description**
queue_id	string (UUID)	Unique identifier of the queue session
clinic_id	string (UUID)	Unique identifier of the clinic
is_active	boolean	Whether the queue session is currently active
currently_serving	object	Details of ticket currently being served
total_waiting	integer	Total number of waiting patients
total_completed_today	integer	Total consultations completed today
avg_wait_time_minutes	integer	Average wait time in minutes
last_updated	string (ISO 8601)	Timestamp of last state update
Sample Response		


```
{
  "queue_id": "q_123e4567-e89b-12d3-a456-426614174000",
  "clinic_id": "clinic_456e7890-e89b-12d3-a456-426614174000",
  "is_active": true,
  "currently_serving": {
    "ticket_number": "A-037",
    "called_at": "2026-10-06T22:01:15Z",
    "counter_number": "Room 3"
```

<table>
<tr><td colspan="3">
<pre>
  },
  "total_waiting": 4,
  "total_completed_today": 36,
  "avg_wait_time_minutes": 12,
  "last_updated": "2026-10-06T22:05:00Z"
}
</pre>
</td></tr>
<tr><td colspan="3">Fail</td></tr>
<tr><td>Status Code</td><td colspan="2">404</td></tr>
<tr><td>Response Type</td><td colspan="2">json</td></tr>
<tr><td>Response Data</td><td>Type</td><td>Description</td></tr>
<tr><td>error_code</td><td>string</td><td>Application-specific error identifier (e.g. QUEUE_NOT_FOUND)</td></tr>
<tr><td>message</td><td>string</td><td>Human-readable description of the error</td></tr>
<tr><td colspan="3">Sample Response</td></tr>
<tr><td colspan="3">
<pre>
{
  "error_code": "QUEUE_NOT_FOUND",
  "message": "The target queue session identifier was not found."
}
</pre>
</td></tr>
</table>

J. Call Next Patient

API Endpoint	/api/v1/queues/{queue_id}/call-next	
method	POST	
Request Parameters	**Type**	**Description**
queue_id	string (UUID)	Required. Path parameter. Unique identifier of the active queue.
room_number	string	Required. Body parameter. Examination room or consultation counter designation.
practitioner_id	string (UUID)	Required. Body parameter. Identifier of

		the attending healthcare provider.
Advances the queue state by calling the next eligible waiting patient and notifying them via SMS/In-App alert.		
Success		
Status Code	200	
Response Type	json	
Response Data	**Type**	**Description**
queue_entry_id	string (UUID)	Unique identifier of called queue entry
ticket_number	string	Ticket number called
patient_name	string	Full name of the called patient
status	string	Queue entry state (e.g. CALLED)
called_at	string (ISO 8601)	Timestamp when patient was called
room_number	string	Assigned consultation room or counter
Sample Response		

{
"queue_entry_id": "qentry_987f6543-e21b-12d3-a456-426614173999",
"ticket_number": "A-038",
"patient_name": "Jane Doe",
"status": "CALLED",
"called_at": "2026-10-06T22:05:56Z",
"room_number": "Exam Room 2B"
}		

Fail		
Status Code	400 / 404 / 409	
Response Type	json	
Response Data	Type	Description
error_code	string	Application-specific error identifier (e.g.

		QUEUE_EMPTY, INVALID_ROOM)
message	string	Human-readable description of the error
Sample Response		

{
"error_code": "QUEUE_EMPTY",
"message": "No waiting patients currently in the queue to call."
}		

K. Update Queue Entry Status

API Endpoint	/api/v1/queues/{queue_id}/entries/{entry_id}	
method	PATCH	
Request Parameters	**Type**	**Description**
queue_id	string (UUID)	Required. Path parameter. Unique identifier of the queue.
entry_id	string (UUID)	Required. Path parameter. Unique identifier of the specific queue entry.
status	string	Required. Body parameter. Target state (IN_CONSULTATION, COMPLETED, NO_SHOW, CANCELLED).
notes	string	Optional. Body parameter. Optional administrative notes regarding status change.
Updates the lifecycle state of a specific patient entry within the queue (e.g., transition to in-consultation, completed, no-show, or cancelled).		
Success		
Status Code	200	

Response Type	json	
Response Data	**Type**	**Description**
queue_entry_id	string (UUID)	Unique identifier of updated queue entry
queue_id	string (UUID)	Unique identifier of the queue
ticket_number	string	Ticket number
status	string	New lifecycle state
updated_at	string (ISO 8601)	Timestamp when status update occurred
notes	string	Administrative notes associated with update

Sample Response


```
{
  "queue_entry_id": "qentry_987f6543-e21b-12d3-a456-426614173999",
  "queue_id": "q_123e4567-e89b-12d3-a456-426614174000",
  "ticket_number": "A-038",
  "status": "COMPLETED",
  "updated_at": "2026-10-06T22:20:10Z",
  "notes": "Consultation finished, follow-up scheduled in 2 weeks."
}
```


Fail

Status Code	400 / 404 / 409	
Response Type	json	
Response Data	Type	Description
error_code	string	Application-specific error identifier (e.g. ENTRY_NOT_FOUND, INVALID_STATE_TRANSITION)
message	string	Human-readable description of the error

Sample Response


```
{
  "error_code": "INVALID_STATE_TRANSITION",
```

"message": "Cannot transition entry status directly from WAITING to COMPLETED."
}